

Perpendicularity of planes

The dihedral angle, its linear angle, and the test that makes two planes perpendicular.

LESSON 40–42

3 × 40 MIN

GEOMETRY 10, §17

EXTENSION

LESSON CLOCK

16'

26'

26'

30'

30'

12

The dihedral angle	16 min
The linear angle	26 min
The test	26 min
Computing dihedral angles	30 min
Practice	30 min
Homework	12 min

LEARNING OBJECTIVES

- Define a dihedral angle and construct its linear angle.
- State and apply the test for perpendicularity of two planes.
- Compute a dihedral angle in a cube and in a pyramid.
- Use the properties of perpendicular planes in proofs.

TEXTBOOK

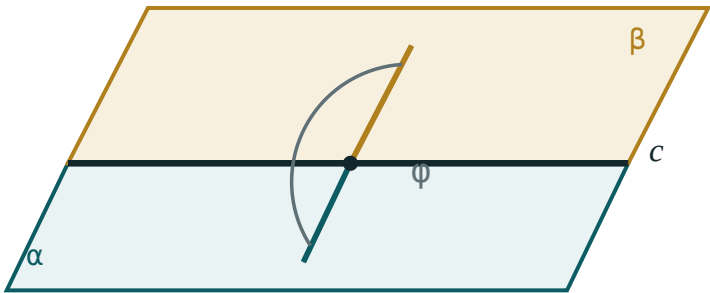
National	pp. 189–206
Cambridge	Core & Extended, pp. 339–344
Workbook	Exercise 17.1

01

Explanation

The dihedral angle

Two half-planes with a common edge divide space; the figure they form is a **dihedral angle**. Its **edge** is the common line; its **faces** are the two half-planes.



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

*The edge, the two faces, and the linear angle
measured between arms perpendicular to the edge.*

THE LINEAR ANGLE

Take any point of the edge and draw, in each face, the ray perpendicular to the edge. The angle between those two rays is the **linear angle**, and it is what the size of the dihedral angle means.

BOTH ARMS PERPENDICULAR TO THE EDGE

Any other pair of rays gives a different, wrong angle. The linear angle does not depend on which point of the edge is chosen — but it does depend on both arms being perpendicular to the edge.

The test for perpendicular planes

DEFINITION AND TEST

Two planes are **perpendicular** when their dihedral angle is right. **Test:** if a plane contains a line perpendicular to another plane, the two planes are perpendicular.

$$\ell \subset \beta \text{ and } \ell \perp \alpha \Rightarrow \beta \perp \alpha$$

In a cube, $AA_1 \perp ABCD$ and $AA_1 \subset ABB_1A_1$, so $ABB_1A_1 \perp ABCD$. Every vertical face of a cube is perpendicular to the base, and one line proves it.

Two useful properties follow:

- If $\alpha \perp \beta$ and a line of α is perpendicular to their common edge, it is perpendicular to β .
- If two intersecting planes are each perpendicular to a third, their line of intersection is perpendicular to that third plane.

Computing dihedral angles

The whole difficulty is the construction of the linear angle. Three reliable routes:

1. **Drop a perpendicular** from a point of one face to the other plane, and join its foot to a point on the edge.
2. **Use the three perpendiculars theorem** to prove the arm you draw really is perpendicular to the edge.
3. **Choose the point on the edge conveniently** — usually a midpoint, so symmetry gives the arms for free.

Example. Cube of edge a : the dihedral angle along AB between the base $ABCD$ and the diagonal plane ABC_1D_1 . The arms are AD in the base and AD_1 in the other face,

both perpendicular to AB . The angle is $\angle DAD_1 = 45^\circ$, since ADD_1 is an isosceles right triangle.

In a pyramid

For a regular square pyramid with base edge $2b$ and height h , two dihedral angles are usually wanted:

ANGLE	ARMS	VALUE
lateral face to base	OM and SM	$\tan \theta = \frac{h}{b}$
lateral edge to base	OA and SA	$\tan \theta = \frac{h}{b\sqrt{2}}$

Base 10 , height 12 : the face angle has $\tan \theta = \frac{12}{5}$, so $\theta \approx 67.4^\circ$; the edge angle has

$\tan \theta = \frac{12}{5\sqrt{2}}$, so $\theta \approx 59.5^\circ$. The face is steeper than the edge — as the picture shows.

02

Worked examples

EXAMPLE 1 Cube of edge a . Find the dihedral angle along AB between $ABCD$ and ABC_1D_1 .

1 $AD \perp AB$ in the base.

2 $AD_1 \perp AB$ in the other face.
 $AB \perp ADD_1A_1$.

3 The linear angle is $\angle DAD_1$.

4 $AD = DD_1 = a$, so it is 45° .

ANSWER

45°

EXAMPLE 2 A regular square pyramid has base 10 and height 12. Find the dihedral angle between a lateral face and the base.

① $OM = 5$, the base apothem.

② $SO = 12$.

③ $\tan \theta = \frac{12}{5} = 2.4$

④ $\theta \approx 67.4^\circ$

ANSWER

$\approx 67.4^\circ$

EXAMPLE 3 Prove that $ABB_1A_1 \perp ABCD$ in a cube.

① $AA_1 \perp ABCD$

By the test of Lesson 33–34.

② $AA_1 \subset ABB_1A_1$

③ By the test for perpendicular planes.

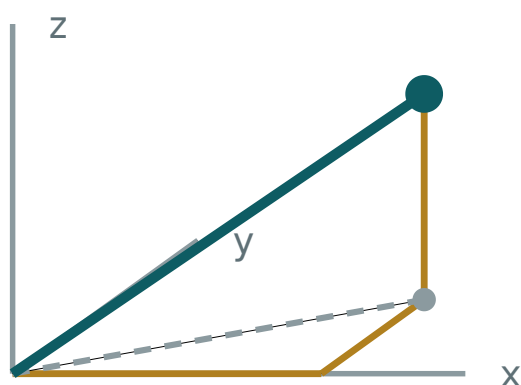
ANSWER

$ABB_1A_1 \perp ABCD$

03 Interactive model

Open a book on the desk: the spine is the edge, the pages the faces, and two pencils give the linear angle.

A dihedral angle



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Edge of a dihedral angle	Ikki yoqli burchak qirrasi	Ребро двугранного угла
Face (half-plane)	Yoq	Грань
Linear angle	Chiziqli burchak	Линейный угол
Perpendicular planes	Perpendikulyar tekisliklar	Перпендикулярные плоскости
Test for perpendicular planes	Tekisliklar perpendikulyarligi alomati	Признак перпендикулярности плоскостей
Right dihedral angle	To'g'ri ikki yoqli burchak	Прямой двугранный угол
Bisector plane	Bissektor tekislik	Биссекторная плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A dihedral angle is measured by:

any angle between the faces

its linear angle

the edge

the normal

The test for perpendicular planes: one contains:

a parallel line

a line perpendicular to the other

two points

the edge

In a cube, ABB_1A_1 and $ABCD$ are:

parallel

perpendicular

at 45°

skew

The dihedral angle along AB between $ABCD$ and ABC_1D_1 is:

Pyramid base 10, height 12: face-to-base angle:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. What is the edge of a dihedral angle?

ANSWER the common line of the two faces

2. How is a dihedral angle measured?

ANSWER by its linear angle

3. Both arms must be $\perp$ to

ANSWER the edge

4. State the test for perpendicular planes

ANSWER one contains a line $\perp$ to the other

5. Is $ABB_1A_1 \perp ABCD$ in a cube?

ANSWER yes

6. Is $ABCD \perp A_1B_1C_1D_1$?

ANSWER no — they are parallel

7. Dihedral angle along AB to ABC_1D_1

ANSWER 45°

Medium · the standard question

7 PROBLEMS

1. Pyramid base 10, height 12: face-to-base angle

ANSWER $\approx 67.4^\circ$

2. Same: edge-to-base angle

ANSWER $\approx 59.5^\circ$

3. Pyramid base 8, height 3: face-to-base angle

ANSWER $\approx 36.9^\circ$

4. Cube: dihedral angle along AB between $ABCD$ and ABB_1A_1

ANSWER 90°

5. Cube: dihedral angle between $ABCD$ and ACC_1A_1

ANSWER 90°

6. Prove $BCC_1B_1 \perp ABCD$

ANSWER $BB_1 \perp ABCD$ and lies in it

7. Regular tetrahedron: angle between a face and the base

ANSWER $\approx 70.5^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : dihedral angle along AC between $ABCD$ and ACD_1

ANSWER $\approx 54.7^\circ$

2. Pyramid base 12, lateral edge 10: dihedral angle of a face with the base

ANSWER $\approx 43.3^\circ$

3. Prove: if two planes are each $\perp$ to a third, their line of intersection is $\perp$ to it

ANSWER It is $\perp$ to both edges

4. A regular tetrahedron of edge a : the dihedral angle between two faces

ANSWER $\arccos \frac{1}{3} \approx 70.5^\circ$

5. Cube: the angle between the planes ABC_1D_1 and A_1B_1CD

ANSWER 90°

6. A prism whose base is a rhombus with a 60° angle: the dihedral angles along the base edges

ANSWER 90° for a right prism

7. Prove that a plane $\perp$ to the edge of a dihedral angle cuts it in its linear angle

ANSWER Both cuts are $\perp$ to the edge

07 Homework

Homework — 6 tasks

Every dihedral answer must name the two arms and say why each is perpendicular to the edge.

1. Define a dihedral angle and explain how its linear angle is constructed.
2. Prove that $DCC_1D_1 \perp ABCD$ in a cube.
3. Cube of edge a : find the dihedral angle along AB between $ABCD$ and ABC_1D_1 .
4. A regular square pyramid has base edge 16 and height 6. Find the dihedral angle between a lateral face and the base.

5. The same pyramid: find the angle between a lateral edge and the base.
6. Prove that if two intersecting planes are each perpendicular to a third, their line of intersection is perpendicular to it.